

Input Port Processing & Destination-Based Forwarding

1 Overview

Input port processing is the first major step in router forwarding. When a packet arrives at a router, the input port must:

- Perform physical- and link-layer reception.
- Decapsulate the incoming frame.
- Look up the correct output port using the forwarding table.
- Optionally queue the packet if the switching fabric is busy.

A conceptual diagram of input processing is shown in Figure 1.

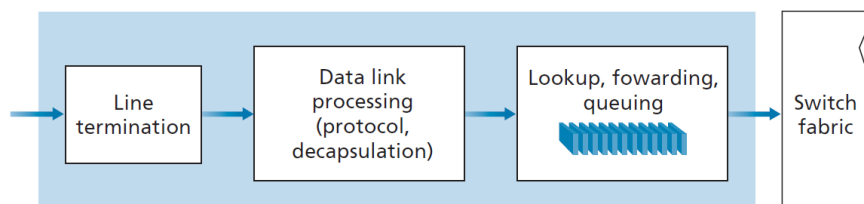


Figure 1: Input port processing: line termination, link-layer processing, lookup, and queuing.

2 Forwarding Table Distribution

Forwarding tables are computed by:

- The **Routing Processor** (traditional designs), or
- A **centralized SDN controller** (SDN architecture).

Shadow Copies on Line Cards

To avoid sending packets to the routing processor for every lookup, each line card receives a *copy* of the forwarding table over an internal bus (e.g., PCI). Thus, lookup occurs locally on each input port at line rate.

This avoids a bottleneck and enables extremely fast packet forwarding.

3 Destination-Based Forwarding

The simplest forwarding method uses only the packet's **destination IP address**.

Because IPv4 addresses are 32-bit, a naive forwarding table would require entries for all 2^{32} (over 4 billion) possible addresses—impractical for real routers.

Instead, routers use **prefix-based forwarding**.

3.1 Example Forwarding Rule

Consider the following destination ranges and their associated interfaces:

Destination Address Range	Interface
11001000 00010111 00010000 00000000 – 11001000 00010111 00010111 11111111	0
11001000 00010111 00011000 00000000 – 11001000 00010111 00011000 11111111	1
11001000 00010111 00011001 00000000 – 11001000 00010111 00011111 11111111	2
Otherwise	3

These ranges can be summarized using prefixes:

Prefix	Interface
11001000 00010111 00010	0
11001000 00010111 00011000	1
11001000 00010111 00011	2
Otherwise	3

Table 1: Prefix-based forwarding table

4 Longest Prefix Match

A key rule for prefix forwarding:

Longest Prefix Match Rule

If multiple prefixes match the destination address, the router selects the entry with the **longest** matching prefix.

Why longest match?

It ensures:

- More specific route entries override broader ones.
- Hierarchical addressing and aggregation are possible.

Example:

Destination:

11001000 00010111 00011000 10101010

Matches:

- Prefix 11001000 00010111 00011 (21 bits)
- Prefix 11001000 00010111 00011000 (24 bits)

Router selects the 24-bit prefix (more specific).

When Destination Address Ranges Are Not Nicely Divided

The example forwarding table shown earlier used very “clean” address blocks that align perfectly with prefix boundaries (e.g., 21-bit or 24-bit prefixes). However, in real networks, destination address ranges are **rarely** this well-behaved.

In practice:

- Organizations may receive multiple disjoint address blocks.
- ISPs often announce prefixes of **different lengths**.
- Address allocations seldom fall on neat binary boundaries.
- Advertised routing prefixes depend on business relationships, not math.

Because of this, ranges like:

11001000 00010111 00010110 00000000 through 11001000 00010111 00011001 11111111

cannot always be represented using a single fixed prefix length.

forwarding table

Destination Address Range	Link Interface
11001000 00010111 00010000 00000000 through 11001000 00010111 00010111 11111111	0
11001000 00010111 00010000 00000100 through 11001000 00010111 00010000 00000111	3
11001000 00010111 00011000 00000000 through 11001000 00010111 00011000 11111111	1
11001000 00010111 00011001 00000000 through 11001000 00010111 00011111 11111111	2
otherwise	3

Figure 2: When Destination Address Ranges Are Not Nicely Divided

5 Fast Lookup Requirements

Routers must perform lookup in nanoseconds to keep up with high link speeds. Thus:

- Lookup must be implemented in hardware.
- Simple linear search is too slow.

Modern routers use:

- Multi-level memory architectures (on-chip DRAM + SRAM cache)
- Tree-based search algorithms
- Ternary Content Addressable Memory (TCAM)

TCAM Advantage

TCAM Lookup

A TCAM can compare the destination IP address against *all entries simultaneously* and return the longest matching prefix in **constant time**.

High-end routers (e.g., Cisco Catalyst 6500/7600) support over a million TCAM entries.

6 Blocking and Queuing

Once the output port is determined:

- The packet attempts to enter the switching fabric.
- If the fabric is temporarily busy (e.g., other packets crossing), the packet is queued at the input.
- A scheduler later transfers queued packets.

Blocking and queuing can occur at:

- Input ports
- Output ports
- Both (depending on router architecture)

7 Other Input-Port Processing Tasks

Besides destination lookup, input ports also perform:

- Checking and updating:
 - IP version
 - Header checksum
 - Time-to-Live (TTL)
- Updating management counters (e.g., number of datagrams received)

8 Match–Action Abstraction

Input processing is an example of a more general concept:

Match + Action Model

Match: Compare header fields with rules.

Action: Forward, drop, rewrite, NAT translation, etc.

Used in:

- Routers (prefix matching)
- Switches (MAC table lookup)
- Firewalls (match → drop)
- NAT devices (match → rewrite)
- SDN devices (multiple header fields tested)

9 Conclusion

Input port processing combines:

- High-speed hardware lookup
- Efficient prefix matching
- Queueing and scheduling
- Generalized match-action operations

It is the fastest, most performance-critical component of a router.